

PARNIKA

Research Engineer — Reinforcement Learning, Robot Imitation Learning & Edge Deployment

· parnika3100@gmail.com · [LinkedIn](#) · [GitHub](#) · [Google Scholar](#) · [Website](#)

SUMMARY

Research engineer focused on reinforcement and imitation learning, alongside real-time deployment of learned models on robotic hardware. Experienced in training learning-based control policies and in taking deep-learning models from research into production. Optimized an autonomous-vehicle semantic-segmentation model to run 3× faster on an NVIDIA Jetson ORIN. M.Sc. in Machine Learning from Mila / Université de Montréal.

TECHNICAL SKILLS

Programming & Tools: Python, Git, Docker, Linux, Singularity

Machine Learning & Computer Vision: PyTorch, NumPy, Vision Transformers (ViT), CLIP

Reinforcement & Imitation Learning: Actor-critic methods, offline RL (TD3+BC), imitation learning (ACT, $\pi 0.5$), policy architecture design, OpenAI Gym, Mujoco

Robotics & Simulation: MuJoCo, teleoperation on SO-101

Deployment & Optimization: TensorRT, edge deployment on NVIDIA Jetson ORIN

Experiment Tracking & Cloud: Weights & Biases, TensorBoard, Microsoft Azure

EDUCATION

M.Sc., Computer Science (Machine Learning) — Université de Montréal

Sep 2023 – Jun 2025

Montréal, QC, Canada

- Relevant coursework: Deep Learning, Robot Learning, Research Internship.

B.Tech., Computer Science & Engineering — Dr. A.P.J. Abdul Kalam Technical University

July 2013 – Jun 2017

Uttar Pradesh, India

EXPERIENCE

Research Intern, REAL Lab — Mila / Université de Montréal

May 2024 – Dec 2024

Montréal, QC, Canada · Advisor: Prof. Glen Berseth

- Designed a transformer-based actor-critic architecture with a novel tokenization scheme to train a single reinforcement-learning policy across multiple robot morphologies simultaneously.
- Conditioned the policy on BERT-encoded URDF (robot-structure) descriptions, enabling generalization to robot embodiments unseen during training, without per-robot retraining. [report](#)

Technologist, Wipro Ltd.

Sep 2022 - Jul 2023

Noida, India

- Optimized a semantic-segmentation model for real-time scene understanding on an autonomous-vehicle platform, targeting the NVIDIA Jetson ORIN edge device.
- Converted the PyTorch model to an INT8-quantized TensorRT engine (calibrated on ~6,000 images), achieving a 3× inference speedup (from ~50 to ~150 fps) while maintaining acceptable accuracy threshold.

Research Collaborator, Indian Institute of Science (IISc)

Oct 2019 - May 2021

Bengaluru, India

- Developed and implemented an actor-critic algorithm to solve constrained optimization for cooperative multi-agent reinforcement learning
- Evaluated and produced results for experiments on 2 benchmark multi-agent environments — Cooperative Navigation and Cooperative Treasure Collection — using an Azure cloud virtual machine.

Research Engineer, Mindtree Ltd.

Nov 2017 - Aug 2022

Bengaluru, India

- Implemented research papers for customer-specific use cases across reinforcement learning, NLP and computer vision.
- Worked on projects including Product Recommendation, Airline Seat Inventory and Dynamic Pricing, and Fine-Grained Visual Classification.

SELECTED PROJECTS

Imitation Learning on a Physical SO-101 Arm

- Collected 192 teleoperated demonstrations with LeRobot for a language-conditioned multitask pick-and-place setup (3 objects × 2 targets), balanced across objects and targets.
- Fine-tuned $\pi 0.5$ with LoRA (openpi) and SmoIVLA (LeRobot) and evaluated manipulation tasks directly on the arm.
- Trained an ACT baseline from scratch on 50 demonstrations of a single-object task: 9/10 task success.

- Results and analysis on this [page](#).

Offline Reinforcement Learning for Vision-Based Manipulation

- Implemented an offline RL (TD3+BC) algorithm for vision-based manipulation on the WidowX arm in simulation, resulting in 3 points better returns than the CQL baseline; also replaced the CNN visual encoder of the baseline with a pretrained ResNet-18 backbone, achieving faster and more stable convergence within 200 epochs. [code report](#)

Multimodal Sarcasm Detection (Vision–Language)

- In multimodal sarcasm detection on MMSD2.0 (paired image–text samples), the label depends on incongruity between modalities rather than on either in isolation.
- Replaced the pre-trained CLIP with LaCLIP(LLM-rewritten caption pretraining) for encoding i/p images and texts, matched LAION-400M corpus and ViT-B/32 architecture to test whether richer text supervision improves cross-modal reasoning. Did NOT find a significant difference in test accuracy across 4 seeds. [Code](#)

Fine Grained Visual Classification

- Learned fine grained visual features for intra-class image classification. Performed FGVC on Stanford-Cars and FGVC-Aircraft datasets with the model architecture (Navigator-Teacher-Scrutinizer). This model learns to propose fine-grained informative regions from the image without the need of bounding boxes as ground truth labels.

PUBLICATION

P. Parnika*, Raghuram Bharadwaj Diddigi*, Sai Koti Reddy Danda* and Shalabh Bhatnagar. "Attention Actor-Critic Algorithm for Multi-Agent Constrained Co-operative Reinforcement Learning." AAMAS 2021. (* equal contribution) [Poster](#) [Code](#)